

 Tianzhong Lan

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Research Interests: Medical Image Analysis, Label-efficient Learning, Multi-modal Medical AI, Trustworthy AI

 **Education**

[1] **Sichuan University** —Ph.D. in Computer Science, Sep 2022–Dec 2026

Advisors: Prof. Zhang Yi and Prof. Min Zhu

[2] **A*STAR, Singapore** —Visiting Ph.D. Researcher, Aug 2025–Jul 2026

Advisors: Dr. Xulei Yang and Dr. Feng Yang

Sichuan University —M.Eng. in Software Engineering, Sep 2019–Jun 2022

North China University of Technology —B.Eng. in Computer Science, Sep 2015–Jun 2019

 Publications

All listed papers are sole first-author works.

[1] **Tianzhong Lan**, Zhang Yi, Xiuyuan Xu, Min Zhu. “GeoCoBox: Box-supervised 3D Tumor Segmentation via Geometric Co-embedding.” AAAI 2026. (Oral)

[2] **Tianzhong Lan**, Weili Jiang, Siyong Yeo, Xulei Yang, Min Zhu. “Wavelet-Driven Cross-Domain Consistency for Mixed-Supervised 3D Tumor Segmentation.” ECCV 2026.

[3] **Tianzhong Lan**, Zhang Yi, Xiuyuan Xu, Min Zhu. “LooBox: Loose-box-supervised 3D Tumor Segmentation with Self-correcting Bidirectional Learning.” ACM MM 2025.

[4] **Tianzhong Lan**, Nan Chen, Zhang Yi, Xiuyuan Xu, Min Zhu. “Domain Generalization for Pulmonary Nodule Detection via Distributionally-Regularized Mamba.” MICCAI 2025.

[5] **Tianzhong Lan**, Fanxin Zeng, Zhang Yi, Xiuyuan Xu, Min Zhu. “ICNoduleNet: Enhancing Pulmonary Nodule Detection Performance on Sharp Kernel CT Imaging.” IEEE Journal of Biomedical and Health Informatics 2024. (JCR Q1)

Manuscripts Under Review:

[6] **Tianzhong Lan**, Weili Jiang, Xulei Yang, Feng Yang, Min Zhu. “Fewer Masks, More Reports: Validation-Guided Closed-Loop Self-Training for 3D Lesion Segmentation.” Under review at AAAI 2027.

[7] **Tianzhong Lan**, Weili Jiang, Feng Yang, Xulei Yang, Min Zhu. “From Invariance to Transport: Flow-Matched Representation Alignment for Continual Lesion Learning.” Under review at IEEE Transactions on Image Processing.

 Patents

[1] **CN112669314A** —Lung Cancer Full-cycle Intelligent Management Imaging Data Platform. Granted; first student inventor.

[2] **CN118212501A** —A Method, System, Device and Medium for Multi-lesion Detection in Lung Cancer. Granted; first student inventor.

Projects

[1] **National Key Project “New Generation AI” 2018AAA0100201**: New-generation Cognitive Neural Network Models, 2019-2023. Developed robust diagnostic models; deployed a lung-cancer full-cycle imaging platform in 10+ tertiary hospitals.

[2] **NSFC Young Scientists Fund 62106163**: Intelligent Detection and Characterization of Lung Cancer, 2022-2024. Developed robust lung cancer detector; deployed a tele-diagnosis system connecting 92 primary hospitals and generating 42,768 clinical reports.

[3] **Tibet S&T Program XZ202402ZY0003**: AI Early-warning Model for High-altitude Pulmonary Hypertension via Multi-omics Digitization, 2024-2026. Focus on label-efficient pulmonary artery segmentation using state-space models.

[4] **Sichuan Key R&D Program 2024YFFK0235**: Clinical Application of Deep Radiomics for Pulmonary Metastasis Prediction, 2024-2026. Focus on deep radiomics modeling and clinical validation for pulmonary metastasis prediction.

[5] **Sichuan Key R&D Program 2023YF0283**: Intelligent Diagnosis of Pulmonary Tuberculosis from Clinical Multimodal Data, 2023–2024. Focus on multimodal learning from CT and clinical text.

Honors & Awards

- National Gold Award, China “Internet+” Innovation and Entrepreneurship Competition, 2022
- China Scholarship Council (CSC) Scholarship, 2025
- PhD Innovation Scholarship, Sichuan University, 2025
- Outstanding Graduate Student, Sichuan University, 2022
- Second-class Scholarship, Sichuan University, 2021, 2024

Skills

Programming: Python, C++ | Deep Learning: PyTorch, nnUNet, Monai | Systems: Linux, Docker
Development: React, Qt5